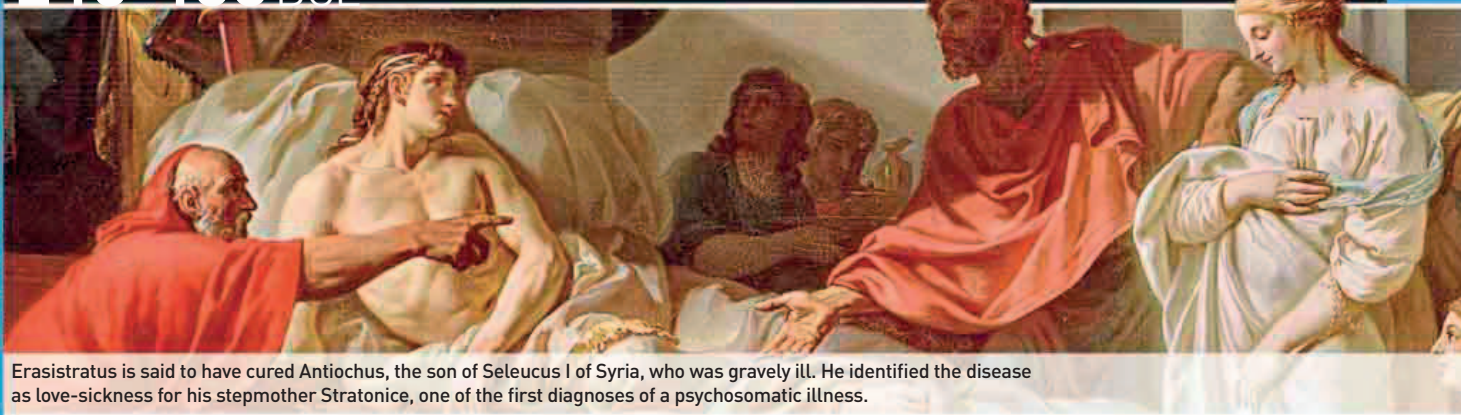
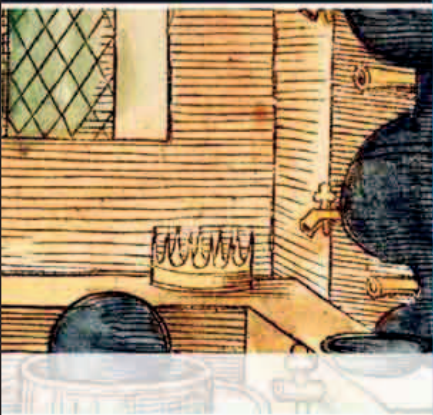
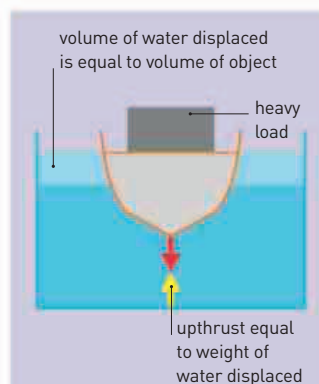




Erasistratus is said to have cured Antiochus, the son of Seleucus I of Syria, who was gravely ill. He identified the disease as love-sickness for his stepmother Stratonice, one of the first diagnoses of a psychosomatic illness.

the Roman conquest of Sicily, in 214 BCE, he was employed by the state to build various machines to defend Syracuse from attack. This included the **Claw of Archimedes**—a type of crane with a huge grappling hook that could capsize enemy ships.



ARCHIMEDES PRINCIPLE

This states that a solid object, partly or wholly immersed in a liquid, has a buoyant force acting on it that is equal to the weight of the fluid it displaces. The relative density of the object can be worked out by dividing the weight of the object by the weight of the displaced liquid. The boat above can support a heavy load because it displaces a lot of water; therefore, the buoyant force supporting it is equally great.

ANATOMY ADVANCED

CONSIDERABLY IN GREECE with the work of **Erasistratus of Cos** (c.304–250 BCE). He developed a theory of **vascular circulation**, in which he said that blood passed through the body in veins, while arteries distributed *pneuma* (air) to vital organs. He also gave an **accurate description of the brain**, including the cerebellum, and distinguished sensory from motor nerves.

Eratosthenes of Cyrene (c.275–195 BCE) made the first map of the world that featured lines of longitude and latitude in around 240 BCE. He also calculated the **dimensions of Earth** by comparing the angles of shadows at noon at Alexandria and Syene in Egypt, which are on roughly the same longitude. He yielded a figure of 250,000 stades—about 29,870 miles (48,070 km)—which is within one percent of the true figure.

Eratosthenes also worked out a simple method of finding prime numbers, known

as the **Sieve of Eratosthenes** (see panel, right).

Greek geometry advanced further in the late 3rd century BCE with the work of **Apollonius of Perga** (c.262–190 BCE), whose major work was entitled *On Conics*. In it he described the properties of the three fundamental **types of conic section**—the ellipse, parabola, and hyperbola. He also developed the **theory of epicycles**—circular orbits rotating around a larger circumference—to refine the theory of the motion of the spheres (see 400–335 BCE).

The Romans found a way of **bonding small stones** to produce **concrete** in the late 2nd century BCE. By adding pozzolana stone (ash from prehistoric volcanoes) to lime, they produced a strong binding mortar. This enabled them to build **stronger and cheaper** monumental buildings. The first structure built



Basilica Maxentius

This early 4th-century concrete Basilica was the largest building in Rome at the time.

	2	3	X	5	X	7	X	X	X
11	X	13	X	X	X	17	X	19	X
X	X	23	X	X	X	X	X	29	X
31	X	X	X	X	X	37	X	X	X
41	X	X	X	X	X	47	X	X	X

SIEVE OF ERATOSTHENES

This is a simple algorithm for finding prime numbers. Starting at 2 without striking it out, strike out all multiples of 2 to the end of the series. Return to the next non-struck out number (3) and without striking it out, strike out every multiple of 3 to the end. Repeat the process; eventually all the non-struck out numbers will be prime.

using concrete was the Porticus Aemilia in Rome in 193 BCE.

Observational astronomy was revolutionized by **Hipparchus of Nicaea** (c.190–120 BCE), who made a **new map of the heavens** that catalogued 850 stars. He invented a new astronomical sighting tool and surveying instrument called the dioptra that was in use until it was replaced by the armillary sphere. Using the dioptra, he discovered the phenomenon of **precession**, by which stars appear to move gradually in relation to the equinoxes.

Hipparchus also calculated the length of the year to be 365.2467 days—very close to the true value.

At this time, the Chinese were busy refining the **production of paper**. The process of soaking and pulping textile rags then drying them out on a screen to produce a fibrous mat for writing on, probably dates from the late 3rd century BCE. Although the **invention of paper** is often ascribed to Cai Lun (50–121), he probably just refined this process and introduced new pulp materials, such as tree bark.

c.250 BCE Erasistratus of Cos distinguishes the cerebellum from the cerebrum

c.240 BCE Eratosthenes of Cyrene measures Earth's dimensions

Before 200 BCE Celts develop single-piece iron tire

c.200 BCE Silver is extracted from lead ore in the Andes

c.200 BCE The kayak is developed in Arctic North America

c.160 BCE Hipparchus of Nicaea describes precession of the equinoxes

c.230 BCE Apollonius of Perga describes the properties of conic sections

c.200 BCE Moche of South America make complex two-piece pottery molds

200 BCE–500 CE Majority of the Nazca geoglyphs are created in Peru

193 BCE The first large concrete building—Porticus Aemilia—is built in Rome